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1 Road extraction from satellite images using deep learning

Keywords: Computer vision, semantic segmentation, multi-model data fusion, weakly-supervised learning and distributed systems

Maps play an important role in life. It was time-consuming and labor-costing for people to get a precise map in the past time. With the development of computer vision and deep learning, road extraction shown in Fig. 1 can be completed by computer and algorithm. Furthermore, with the increment of data recently, a computer-aided road extraction system can be built. Our work focuses on improving the performance of road extraction by combining satellite images and GPS tracks.

In CVPR 2018 workshop, Facebook Inc. organized a competition about road extraction [2]. In this competition, DLinkNet won its championship with an IoU about 60% [14]. It is obvious to find the road would be hidden by the plants and building shades if we only use satellite images shown in Fig. 2, which is even hard for people to recognize the road, to say nothing to computer vision algorithms. Also, different colors and structures will bring difficulties in finishing this task. Roads are different in developed and developing countries. To overcome these shortcomings, we considered combining GPS tracks and satellite images.



Figure 1: Road Extraction Illustration

How to use GPS information in the computer vision system is challenging. In addition to the three channels of RGB as the input of the network, we rendered the GPS points on the image as another channel shown in Fig. 2. Besides latitude and longitude, we also tried the direction of vehicles and their speeds. However, the performance is not as good as being rendered according to the amount.

Like road structure in different areas, the quality of GPS also varies. In our experiments, we found that the network relies on GPS very much because GPS is a kind of structured data. To avoid overfitting, we tried to change the amount and the precision of GPS. We proposed sub-sampling, sub-resolution and random omission shown in Fig. 3. We found that our method can overcome the different GPS conditions effectively as shown in Fig. 5. Besides GPS data, we also proposed a 1D de-convolution kernel in the decoder part to make the network produce the road better. The framework of our network and the details of the 1D de-convolution kernel are shown in Fig. 4.

After acceptance of CVPR 2019 [11], we considered improving our results. Inspired by Facebook's two papers about weakly-supervised learning [8, 9], we are trying to build a system using global satellite images and OpenStreetMap (OSM) data. We built a cluster with 50 GPUs for distributed training. The distributed framework we used is PyTorch 1.4.0 and we installed these 50 GPUs in 5 separate devices. We wanted to use the images with rough labels to find the edge of

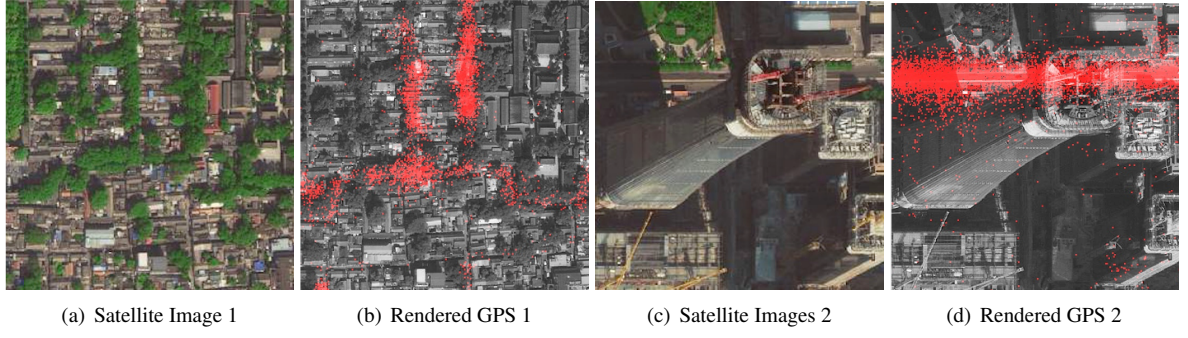


Figure 2: The illustration which GPS can help satellite images. For (a) and (b), plants make the road unobserved. For (c) and (d), the shade of building makes the road unobserved.

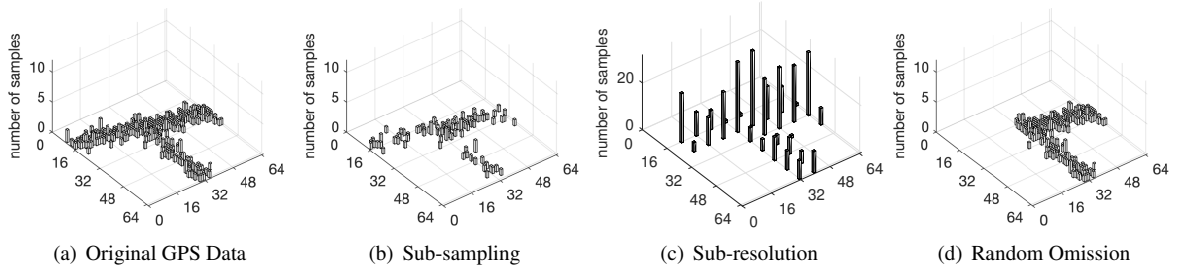


Figure 3: The methods we used for GPS data augmentation

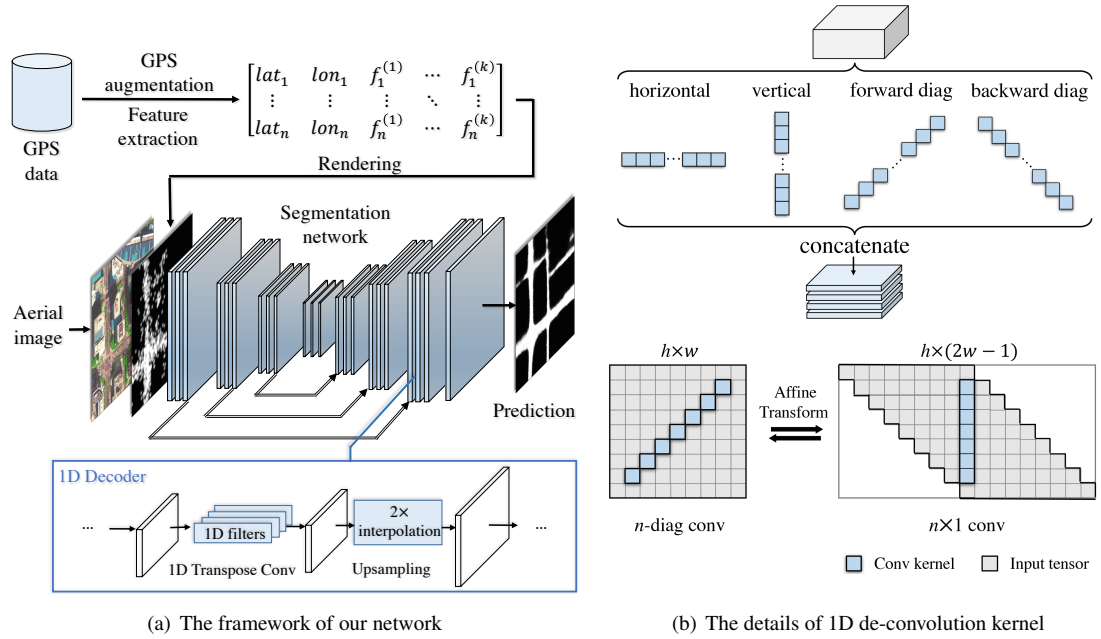


Figure 4: The framework of our network and the details of 1D de-convolution kernel

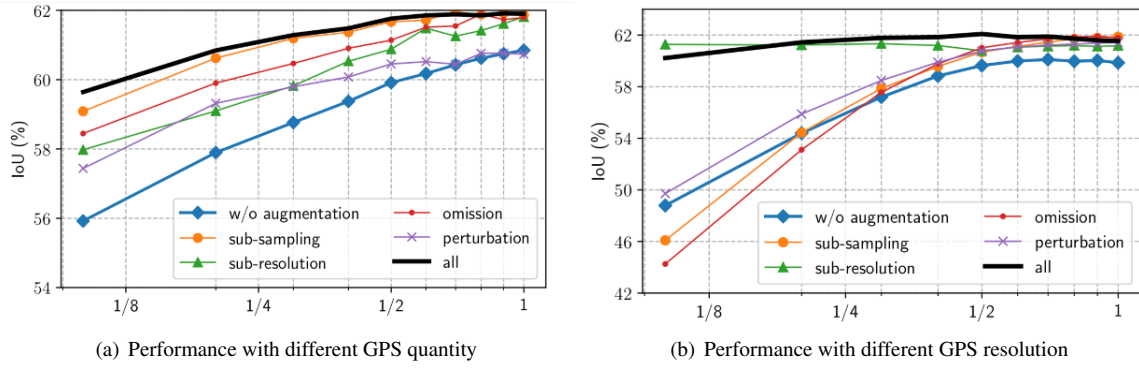


Figure 5: Results from difference in GPS precision and quantity

weakly-supervised learning. Fig. 6 shows the preliminary result in Beijing. It can be found that our model can give a result which is not bad. We are composing a paper for ECCV 2020 submission.

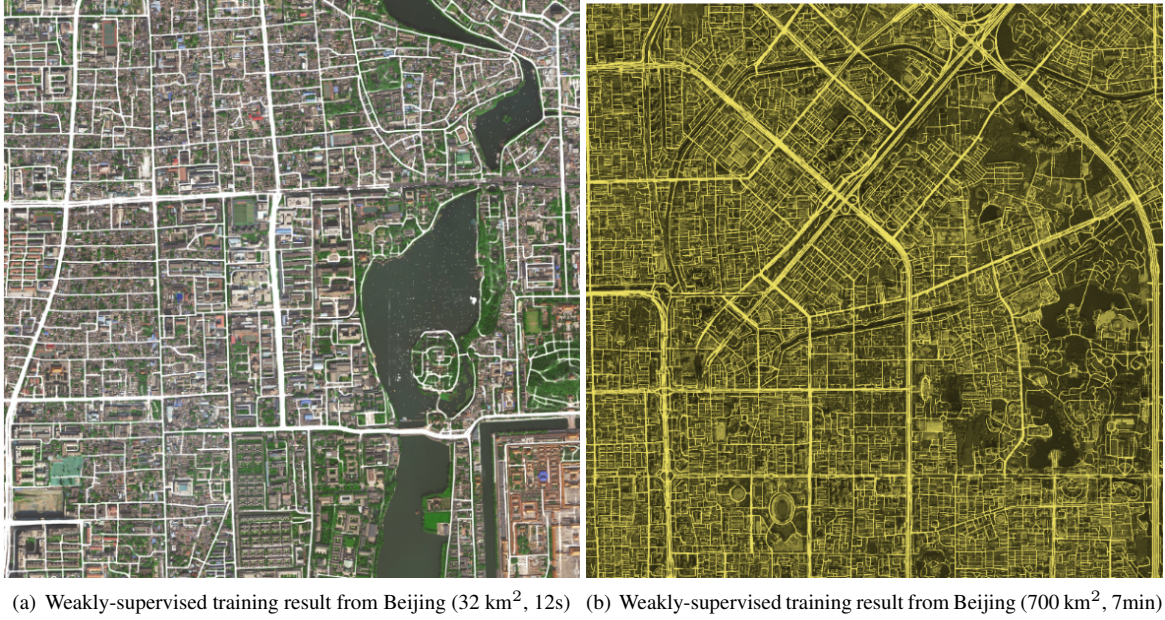


Figure 6: Preliminary results from weakly-supervised learning in Beijing

2 Imbalance classification using density peak clustering (DP) and support vector machine (SVM)

Keywords: Imbalance classification, density peak clustering and support vector machine (SVM)

Imbalance classification is a common problem in machine learning. To deal with it, we used density peak clustering [10] and adaptive density peak detection [12] in multi-class and binary imbalance classification. The classifier we used in both tasks is SVM [1]. The goal of SVM is to find a hyper-plane to separate two classes with the maximum margin, which is equivalent to

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^l \alpha_i - \frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l \alpha_i \alpha_j y_i y_j \kappa(x, y) \\ \text{s.t.} \quad & \sum_{i=1}^l \alpha_i y_i = 0, \alpha_i \leq 0, i = 1, 2, \dots, l. \end{aligned} \quad (1)$$

The main idea of clustering is that the difference among data points in the same cluster is as small as possible while the difference among points in different clusters is as large as possible. Clustering by fast search and find of density peaks (DP) [10] uses Euclidean distance to evaluate the difference between points. DP defines two parameters: local density ρ_i of point i and the distance between point i and point j with higher density δ_i . A cluster center should have both large ρ and δ . However, DP fails to answer how to set the threshold of ρ and δ in theory and the selection of the cluster center is from user's input manually. This works for multiple clusters because the number of cluster centers is known as $n - 1$ where n is the number of class of the dataset. In [5], we tried to combine DP and SVM together to solve the imbalance classification problem and the method can solve the problem in some way. The algorithm is shown below.

Algorithm 1: Multi-class imbalance classification using DP and SVM

Input: Training set $D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_l, y_l)\}, y \in \{1, 2, \dots, k\}$

Output: Classification Results

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1 for  $i = 1; i \leq k$  do
2    $D = \{X_i, \bar{X}_i\}$ , where  $X_i$  is the data with label  $i$  and  $\bar{X}_i$  is the rest of dataset;
3   if  $(|X_i| \leq |\bar{X}_i|)$  then
4      $M = \bar{X}_i$ ;
5      $m = X_i$ ;
6   else
7      $m = \bar{X}_i$ ;
8      $M = X_i$ ;
9   Apply DP on  $M$  and the number of cluster center is  $k - 1$  and  $M$  can be  $M = \{M_1, M_2, \dots, M_{k-1}\}$ , where
      $M_n (n = 1, 2, \dots, k - 1)$  is the cluster after DP;
10  Under-sample  $M_n$  according to  $|M'_n| = (|M_n| \cdot |m|) / \sum_{j=1}^{k-1} |M_j|$ ;
11   $M' = \{M'_1, M'_2, \dots, M'_{k-1}\}$ , where  $|M'| \approx |m|$ ;
12   $\text{results}[i] = \text{SVM}(m, M')$ ;
13 return results;
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Compared with binary classification, there are fewer works about multi-class imbalance classification. To solve the multi-class classification problem, SVM has two strategies which are one-vs-one (O) and one-vs-rest (R). We compared all the methods to test our idea. We used 5 datasets in the experiments. The results are shown in Fig. 7.

ADPclust [12] is a modified method of DP which can set the parameter in a more adaptive way by using a continuous density calculation method and a double-check process to guarantee the selection of cluster centers is the best. The framework of the method is shown in Fig. 8. We tested our method on 10 datasets and the results are shown in Fig. 9.

In addition, To clarify the difference between our methods and other methods, we applied t -test on our methods both in binary [6] and multi-class [5]. The results also prove our methods' uniqueness.

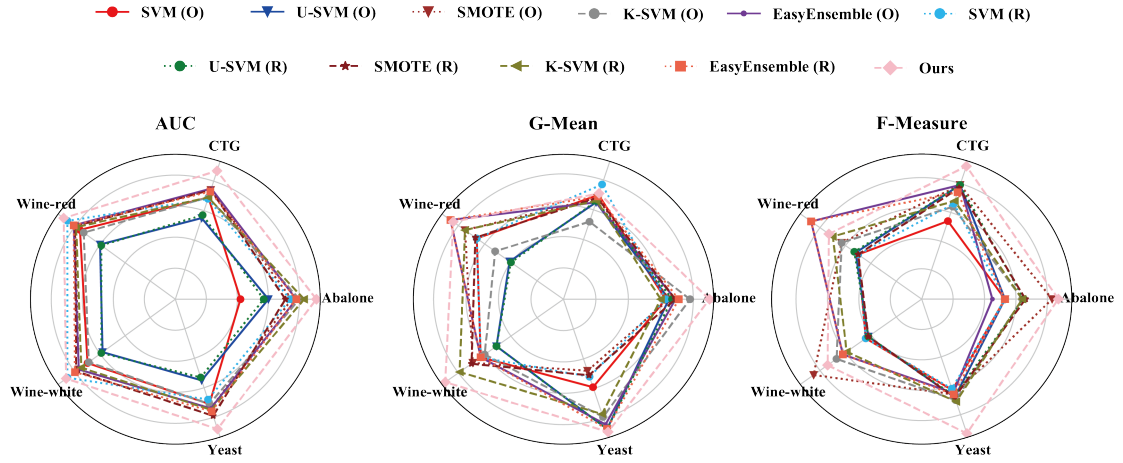


Figure 7: The results of multi-class classification

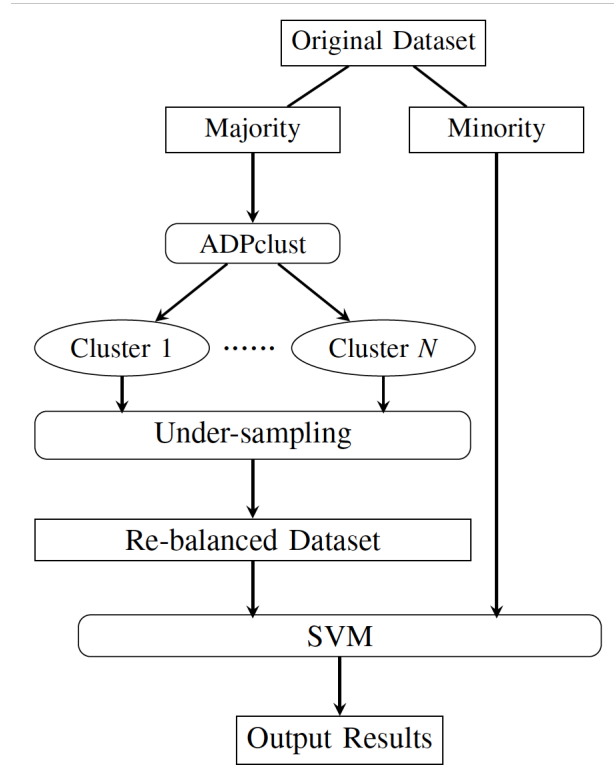


Figure 8: The framework of binary classification

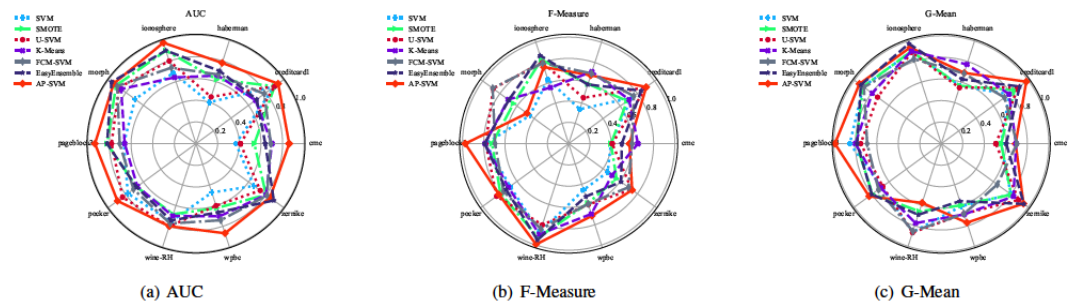


Figure 9: The results of binary classification

3 Internet addiction (IA) detection using SVM and multiple questionnaire data

Keywords: IA detection, SVM, a multiple-questionnaire method and student's t test

IA is a serious illness around the world. The traditional method to detect IA is using questionnaires. However, the questionnaire method is subjective. In practice, only one single questionnaire is used to detect IA, which is hard to judge the relationship between IA and other characteristic sub-fields, like Openness, Extraversion, Conscientiousness, Agreeableness, Neuroticism, etc. To overcome the existing shortcomings and find a quantitative relationship, we used age, gender and 9 sub-fields as features and Chen Internet Addiction Scale (CIAS) as the label to find out dependency between IAD and the characteristic sub-fields. These 9 sub-fields including Openness, Extraversion, Conscientiousness, Agreeableness, Neuroticism, Attention, Motor, Non-planning and Self Control are in 3 questionnaires. These 3 questionnaires are Brief Self Control Scale (BSCS), the 11th version of Barratt Impulsiveness Scale (BIS-11), Chinese Big Five Personality Inventory (CBF-PI). We collected data from 2397 Chinese college students. Because the problem is binary classification, the classifier we used is SVM. To improve the performance of SVM and generalization, we applied student's t test on the features and 10-fold cross-validation on the dataset. To our best knowledge, [13] is similar to our idea, so we also tested fuzzy neural network (FNN). The framework in this experiment is shown in Fig. 10. To find the best parameters of SVM, we used a grid search. Finally, we found that our classifier (C-SVM with g -step 37) can give a 96.06% accuracy. We also found that BIS-11, BSCS, Motor, Neuroticism, Non-planning and Conscientiousness are shown as a better detection feature combination of IA detection. Our finding points a new direction of IA detection and is published in [3, 4, 7]

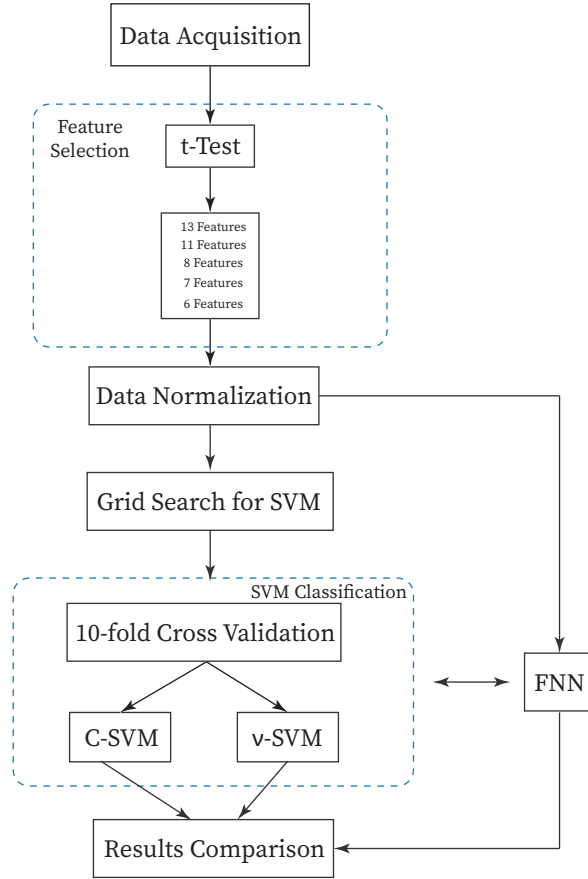


Figure 10: The framework we used in IA detection

4 5-stage pipeline CPU

Keywords: MIPS, 5-stage pipeline CPU, FPGA Programming and Verilog

In the *Computer Organization & Architecture* course, we are asked to finish a 5-stage pipeline CPU using a field programmable gate array (FPGA) and Verilog. The FPGA is Xilinx and the software is Vivado. It is important to control the clock, define models and re-use these models. I defined every model at the beginning of programming, like operator (adder, multiplier, divider), registers, etc. For example, we should define a ROM to store all the instruction code first and drive the clock to fetch the code and push it into the next part of the CPU. When I was trying to implement an instruction, I first searched the format of the code to find out the meaning of every bit range. Table 1. shows the example of add instruction. The operation code is 0000 00. The first and second parameters are ss sss and t tttt. The destination is dddd d. If the CPU reads text add, the operation code would be given 0000 00 and the operator would be set as adder. The input would be set as ss sss and t tttt. After the operation, dddd d would be the result. The other instructions are in the same way. I totally implemented 54 instructions.

Description:	Adds two registers and stores the result in a register
Operation:	\$d = \$s + \$t; advance_pc (4);
Syntax:	dd \$d, \$s, \$t
Encoding:	0000 00ss sst tttt dddd d000 0010 0000

Table 1: MIPS add instruction description

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